

zic1 (XENLA, UniProt O73689) — Hypothesis Review

Focus: proposed GO term "neural plate border formation";
seed = pax3+zic1 border co-expression (scRNA)

Executive Judgment

Verdict: Supported (core claim) / Partially supported (specific details), with important dynamic qualifiers.

The seed hypothesis's central biology — that at the *Xenopus* neural plate border (NPB) pax3 and zic1 are co-expressed in the same cells, that the pax3+zic1 double-positive population is the one that progresses to neural crest (snai2/sox10/foxd3), and that zic1-only cells adopt preplacodal/anterior-neural fate — is **strongly and directly supported by causal, functional primary literature in *Xenopus***

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P 15843410

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P 17409353

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This is stronger than the correlative single-cell co-expression test the seed proposes.

Importantly, the specific single-cell experiment the seed asks for has effectively **already been performed in the correct organism at higher spatial fidelity** by quantitative HCR in situ hybridization (Montequin & LaBonne,

P 41718037

 / preprint

P 41256729

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That work **confirms** co-expression and the double-positive→ crest logic but **qualifies** the static snapshot model: pax3/zic1 onset is broad, non-uniform and partially overlapping; relative (not merely binary) pax3:zic1 levels tune crest-gene choice (snai2 vs sox8); and at later stages NPB

factors are **inversely** correlated with crest identity because pax3/zic1 are downregulated as crest cells commit. A naïve "double-positive cells = crest" readout from a whole-embryo scRNA atlas (Briggs et al. 2018) is therefore liable to stage-dependent misinterpretation.

For curation: the underlying gene function (zic1 is a core NPB transcription factor whose combinatorial activity specifies NPB derivatives) is well supported. **A live GO/QuickGO check (this iteration) shows this biology is already annotated** — zic1 (O73689) already carries GO:0014034 *neural crest cell fate commitment* (IMP), GO:0014033 *neural crest cell differentiation* (IMP), GO:0001840 *neural plate development* (IMP) and GO:0014029 *neural crest formation* (IEA). There is **no dedicated "neural plate border" GO term**, but GO:0014029's own definition literally describes formation of the border region. So the proposed new term "neural plate border formation" is biologically apt yet **largely redundant with an existing term's definition**; its real value is as an **ontology-structure refinement** (splitting border-territory formation from crest formation), not as a missing gene-level annotation. See GO Curation Implications.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
P 15843410 (Sato, Sasai & Sasai 2005)	Mutant/mis-expression (gain+loss of function)	Supports	pax3+zic1 co-expression specifies crest	Co-expressed in presumptive crest before Foxd3/Slug; co-misexpression induces ectopic crest (Wnt-dependent); either alone only expands crest in dorsolateral ectoderm; co-presence necessary to initiate crest	Xenopus gastrula ectoderm; whole embryo + animal cap	High; evidence not in single quantitative
P 17409353 (Hong & Saint-Jeannet 2007)	Mutant phenotype (gain + knockdown)	Supports	double-positive→crest; zic1-only→preplacodal	At end of gastrulation pax3+zic1 coexpressed in crest-forming region; Pax3-only→hatching gland, Zic1-only→preplacodal, combined→crest; level manipulation shifts fates	Xenopus NPB; whole embryo + animal explants	High; essential three the se
P 41718037 / P 41256729 (Montequin & LaBonne 2025/26)	Direct quantitative assay (HCR in situ, single-cell resolution)	Supports + Qualifies	Per-cell co-expression dynamics of pax3/zic1 vs crest genes	Broad, partially overlapping onset with AP/ML biases; relative pax3:zic1 intensity predicts snai2 vs sox8 (confirmed); later stages show	Xenopus NPB, neurula time course	High; test. S seed's mode overs

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
				inverse correlation as pax3/zic1 are downregulated when crest identity emerges		
P 30012125 (Maharana & Schlosser 2018)	Gain/loss-of-function GRN	Supports	zic1 required for both preplacodal and crest	Zic1 (dorsal/neural factor) required context-dependently for both preplacodal ectoderm and neural crest; cross-regulated by Six1/Eya1	Xenopus laevis NPB GRN	High; as a b-speci crest-
P 29852131 (Pla & Monsoro-Burq 2018, review)	Review/synthesis	Qualifies (orientation)	NPB patterning & crest induction	Places pax3/zic1 within the neural border GRN; border is multipotent (crest + dorsal neural tube + nonneural + anterior placodes)	Vertebrate NPB	Review orien
P 34638777 (Bellchambers et al. 2021)	Mutant phenotype (mouse)	Qualifies / competing	Conservation of ZIC1 border role	Calls ZIC1 a "core component" of the vertebrate crest GRN but notes mammalian crest formation "remains open to question"; SUMOylation modulates ZIC activity	Mouse embryo	Medi organ differ Xenop may n uncha mam

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
P 40939754 (Godden et al. 2025)	Mutant phenotype + RNA-seq of NPB/NC explants	Supports	zic1/pax3 are upstream NPB/NC markers	miR-196a knockdown perturbs sox2/3, zic1/3, pax3, sox10, snail2; zic1/3 + pax3 are the border markers balancing neural vs crest fate	Xenopus laevis dorsal ectoderm	Medi... reinforce NPB n... show... borde
P 32891623 (Neilson et al. 2020)	Mutant phenotype	Qualifies (paralog)	zic2 at the border	Mcrs1 knockdown expands zic1 and zic2 (neural plate + neural crest genes)	Xenopus laevis	Medi... zic2 (borde
P 32640092 (Rahman et al. 2020)	Mutant phenotype	Supports (function)	zic1 patterns neural plate/crest	zic1 regulates neural plate patterning, crest formation, cerebellar development via downstream prdm12	Xenopus	Medi... corro... zic1's patter
P 16871625 (Fujimi et al. 2006)	Expression + mis-expression (all 5 paralogs)	Supports (paralog claim)	zic2-5 border co-expression	zic4 "detected mainly in the neural plate border, dorsal neural tube... similar to zic1"; all 5 zic genes cooperatively regulate neural/crest development "despite significantly	X. laevis neurula	Medi... doma... expre... single... ident

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
				diverged expression profiles and functions" (two functional groups zic1/2/3 vs zic4/5)		
P 9655809 (Kuo et al. 1998)	Expression + gain of function	Supports	zic(Opl) in border/dorsal neural/crest	Opl/Zic expressed throughout presumptive neural plate then restricted to dorsal neural tube and neural crest; activated form induces crest and dorsal neural markers	X. laevis	Mediocre expression dynamics
P 29442320 (Merzdorf & Forecki 2018)	Review/database	Qualifies (orientation)	zic family roles	Synthesis: zic factors drive early neural patterning and neural crest specification; act as both activators/repressors	Xenopus	Review orientation

GO Curation Implications (leads — require curator verification)

Verified against the live GO ontology (EBI OLS) and QuickGO on 2026-10-09 — direct queries, see Provenance.

Key ontology finding: GO has **no dedicated "neural plate border" term** (searches for "neural plate border", "...specification", "...formation" return none). The closest existing terms are **GO:0014029 neural crest formation**, the neural-plate-regionalization family (GO:0060896

neural plate pattern specification; GO:0021998 mediolateral; GO:0021999 A/P; GO:0060897 regionalization), and GO:0021502 neural fold elevation formation. Critically, **GO:0014029's own definition already is the border**: *"The formation of the specialized region of ectoderm between the neural ectoderm (neural plate) and non-neural ectoderm. The neural crest gives rise to the neural crest cells that migrate away from this region..."* So the existing term labelled "neural crest formation" is biologically a **neural-plate-border-formation** term.

Key annotation finding: *Xenopus zic1* (O73689) is **already annotated** to the relevant processes, several with experimental (IMP) evidence:

GO ID	Term	Evidence	Relevance to seed
GO:0014029	neural crest formation (def = border-region formation)	IEA	Already captures the "border formation" concept
GO:0014034	neural crest cell fate commitment	IMP	Core crest-specification role (experimental)
GO:0014033	neural crest cell differentiation	IMP	Crest arm of the border
GO:0001840	neural plate development	IMP	Neural-plate/border context
GO:0021904	dorsal/ventral neural tube patterning	IMP	Downstream neural patterning
GO:0030177	positive regulation of Wnt signaling pathway	IGI	Matches Wnt-dependence of crest induction
GO:0045944 / GO:0045893	positive regulation of transcription (Pol II / DNA-templated)	IMP	MF-level TF activity

Curation decision table (leads — require curator verification):

Option	Assessment
Add new term "neural plate border formation" and annotate zic1	Biologically apt, but substantially redundant with existing GO:0014029 whose definition is border formation. Justifiable only as an ontology-structure request to split "border territory establishment" from "neural crest formation" (crest-centric label mismatches its border-level definition). Treat as an ontology-development lead, not a new gene fact.
Retain/rely on existing GO:0014029 (neural crest formation)	Already present (IEA). Could be upgraded to experimental evidence given P 15843410 / 17409353, but note zic1 is combinatorial (needs pax3 + Wnt) and also drives the preplacodal arm, so a crest-only label under-describes zic1.
Rely on existing GO:0014034 / GO:0014033 (IMP)	These already capture zic1's experimentally-demonstrated crest-specification role — the seed's core biology is not unannotated .
Border-level framing (preferred biologically)	zic1 is required for both crest and preplacodal border derivatives P 17409353); P 30012125). A true neural-plate-border parent term would describe zic1 more accurately than any crest-only term — this is the strongest argument <i>for</i> the proposed new term as an ontology improvement.

- Aspect: this is a **Biological Process** matter (plus the already-annotated MF transcription-factor terms); not a CC claim.
- **Avoid "protein binding"** as the curation outcome — specific developmental BP terms are well supported.
- Recommended qualifiers if any annotation is added/upgraded: experimental evidence (IMP/IGI) from

P 15843410

and

P 17409353

taxon = Xenopus; annotate with awareness that the function is **combinatorial with pax3** and **Wnt-dependent**, and that zic1 alone biases to preplacodal fate.

- **Bottom line for the curator:** the seed's underlying biology is real and already represented in *zic1*'s GO record (GO:0014034/GO:0014033 IMP; GO:0014029 IEA). The proposed new term adds value mainly as an **ontology refinement** (separating border formation from crest formation), not as a missing gene annotation.

Mechanistic Scope

- **Immediate molecular function:** *zic1* is a zinc-finger (ZIC/GLI family) transcription factor acting at the NPB. Its direct, cell-autonomous activity is transcriptional regulation of border/fate genes.
- **Direct cellular process:** specification of neural-plate-border cell fates — with *pax3* (+ Wnt) it specifies neural crest; alone it biases toward preplacodal ectoderm.
- **Downstream / not the direct activity being annotated:** overt neural crest EMT/migration, craniofacial and placodal derivatives, cerebellar patterning, *prdm12* induction — these are downstream consequences, not the immediate molecular function. Curation of the proposed term should rest on the specification activity, not on the distal phenotypes.

Conflicts and Alternatives

1. **Dynamics vs static snapshot (most important).** The seed frames double-positive cells as "the" crest progenitors across stages. Montequin & LaBonne

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P 41718037
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show *pax3/zic1* are **downregulated** as crest identity consolidates, producing an **inverse** NPB-vs-crest correlation at later stages. A Briggs-atlas per-cell co-expression readout could therefore *under-count* committed crest cells at later stages and should be stage-stratified.

2. ***zic1* is not crest-specific.** *zic1* is required for preplacodal fate too

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P 17409353
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P 30012125
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Treating *zic1* primarily as a crest gene would be too narrow.

3. **Paralog claim now supported at the domain level (Iteration 3).** The seed asserts *zic2*, *zic3*, *zic4*, *zic5* are also in the *pax3+zic1+* border population. *zic2*

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P 32891623
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zic3

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P 40939754),

and critically **zic4**

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P 16871625),

"**neural plate border, dorsal neural tube... similar to zic1**") are documented at the Xenopus border; all five zic paralogs co-regulate neural/crest development

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P 16871625).

Caveat: profiles are "significantly diverged" with two functional groups (zic1/2/3 vs zic4/5), so strict *per-cell* co-expression of all five in the identical double-positive cells is an oversimplification not formally shown at single-cell resolution. Potential paralog cross-reactivity/mis-mapping is a real caveat for any probe- or read-mapping- based quantification (X. laevis is allotetraploid with L/S homeologs; zic paralogs are closely related and zic1/zic4 are chromosomally adjacent).

4. **Organism specificity.** The combinatorial Pax3+Zic1 crest model is best established in Xenopus; its mammalian conservation is explicitly "open to question"

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P 34638777).

This matters only if the annotation is projected beyond Xenopus.

Knowledge Gaps

• Briggs 2018

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P 29700227)

per-cell quantification not run here. The whole-embryo X. tropicalis scRNA atlas was not downloaded/analyzed in this review (dataset retrieval/compute not performed; not necessary given the superior targeted Xenopus HCR data already available). Gap matters because the seed specifically names it; resolving it would require loading the atlas, identifying NPB/early-crest clusters, and computing stage- stratified per-cell co-expression of pax3, zic1–5 and snai2/sox10/foxd3 — explicitly separating early (double-positive→crest) from late (inverse-correlation) stages.

- **zic paralog claim — RESOLVED at domain level (Iteration 3).** zic4 is explicitly a neural-plate-border gene like zic1

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P 16871625);

zic2

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P 32891623),

zic3

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P 40939754)

documented at the border; all five zic genes co-regulate neural/crest development.

Remaining gap: strict *per-cell* co-expression of all five (esp. zic5) in the identical double-positive cells is not shown at single-cell resolution, and profiles are "significantly diverged."

- **Whether GO already contains a suitable term — RESOLVED this iteration** (direct OLS/QuickGO query): no dedicated neural-plate-border term exists, but GO:0014029's definition already describes border formation, and zic1 is already annotated to GO:0014029/GO:0014034/GO:0014033. The remaining gap is an **ontology-design question** (should "border formation" be split from "crest formation"?) for the GO consortium, not a missing gene annotation.

Discriminating Tests

1. Re-analyze Briggs et al. 2018

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P 29700227)

stage-by-stage: per-cell co-expression of pax3 × zic1–5 vs snai2/sox10/foxd3, testing the prediction that double-positive fraction co-expressing crest specifiers rises early then falls (inverse correlation) as in

P 41718037.

2. Lineage tracing / live imaging of pax3+zic1+ cells to confirm they are the cells that become crest (co- expression alone cannot establish progression).
3. Quantify **relative** pax3:zic1 ratio per cell against snai2 vs sox8 to test the dose-tuning model.
4. Paralog-resolved probes/reads (handle L/S homeologs and zic2–5) to test the multi-zic co-expression claim.

Curation Leads (require curator verification)

- **References to attach:**

[P 15843410](#)

and

[P 17409353](#)

(primary functional support for the border- specification/combinatorial model);

[P 41718037](#)

(direct quantitative single-cell-resolution test + dynamic qualifier); supporting:

[P 30012125](#),

[P 40939754](#),

[P 32891623](#).

- **Candidate GO action:** support a BP annotation for *zic1* at the neural plate border. Prefer an existing GO BP term if one covers "neural plate border / neural crest specification"; if the proposed new term "neural plate border formation" is adopted, consider wording it as **border cell-fate specification** and apply combinatorial/ contributes_to framing (*zic1* + *pax3* + Wnt), not sole-driver.

- **Snippet to verify**

([P 17409353](#)):

"At the end of gastrulation, Pax3 and Zic1 are coexpressed in the neural crest forming region ... Zic1 is detected in the preplacodal ectoderm ... their combined activity is essential to specify the neural crest."

- **Snippet to verify**

([P 41718037](#)):

"later stages display an inverse correlation between neural crest and neural plate border factors, suggesting that *pax3* and *zic1* initially promote neural crest gene activation but are downregulated as neural crest identity emerges."

- **Suggested curator questions:** (a) Does GO already have a neural-plate-border term? (b) Should the annotation be crest-specific or border-level (recommend border-level for

accuracy)? (c) Is the combinatorial/Wnt-dependent nature captured? (d) Keep annotation taxon-scoped to *Xenopus* given mammalian uncertainty.

- **Suggested experiment:** the stage-stratified Briggs re-analysis above, to directly close the seed's named gap.

Provenance (analyses actually run)

- **GO ontology check (EBI OLS4 API, 2026-10-09):** searched GO for "neural plate border", "neural plate border specification/formation", "neural crest formation", "neural plate pattern", "roof plate". Result: no dedicated neural-plate-border term; retrieved definitions of GO:0014029, GO:0060896, GO:0021998/0021999/0060897, GO:0021502. GO:0014029 definition verbatim = "The formation of the specialized region of ectoderm between the neural ectoderm (neural plate) and non-neural ectoderm...".
- **Existing-annotation check (QuickGO API):** retrieved BP annotations for *zic1* O73689 (22 BP terms incl. GO:0014029 IEA, GO:0014034 IMP, GO:0014033 IMP, GO:0001840 IMP, GO:0021904 IMP, GO:0030177 IGI, GO:0045944/ 0045893 IMP) and human *ZIC1* Q15915 (20 BP terms) for cross-species comparison.
- These queries are reproducible with the `requests` calls shown in the executed code cells this iteration. No values were hand-entered; term labels were resolved programmatically from OLS.

Limitations

This review is literature- and reasoning-based (plus the live GO/QuickGO queries above). The named scRNA atlas (Briggs 2018) was not computationally re-analyzed in this run; conclusions rest on causal functional studies and a direct quantitative in-situ study in *Xenopus* that supersede a pure co-expression correlation. No results were fabricated; where a check was not performed it is stated plainly.